

The Holographic Principle: 0/0 of Information

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The Removable Singularity Framework

Abstract

We show that the holographic principle is the 0/0 of information: all information in a volume of space can be encoded on its boundary. The Bekenstein-Hawking entropy $S = A/(4G_N)$ IS a 0/0: information at the horizon is neither destroyed nor preserved. The Ryu-Takayanagi formula connects entanglement to geometry. AdS/CFT shows the 3D world is a hologram of 2D information. This connects black holes, entanglement, consciousness, RMT, networks, and the cosmic web.

1. Bekenstein-Hawking Entropy

$$S = A / (4 * G_N)$$

The entropy of a black hole is proportional to its AREA, not volume. This is counterintuitive: more volume should mean more entropy. But black holes have $S \sim r^2$, not $S \sim r^3$. This is the holographic 0/0: information at the boundary encodes everything.

2. Ryu-Takayanagi Formula

$$S_A = \text{Area}(\gamma_A) / (4 * G_N)$$

The entanglement entropy of a region equals the area of the minimal surface dividing it, divided by $4G_N$. This connects quantum information to spacetime geometry. Entanglement IS geometry.

3. Information Paradox

Information falls into a black hole. Is it destroyed (violating unitarity)? Is it preserved (violating no-cloning)? At the horizon: 0/0. The Page curve shows information IS preserved: before t_{Page} , radiation is thermal; after t_{Page} , radiation contains information. Unitarity is maintained.

4. AdS/CFT Correspondence

$$\text{D-dim gravity} = (\text{D-1-dim CFT})$$

The AdS/CFT correspondence states that D-dimensional gravity is equivalent to (D-1)-dimensional conformal field theory. The bulk (interior) is dual to the boundary (surface). The 3D world is a HOLOGRAM of 2D information.

5. Bekenstein Bound

$$I \leq A / (4 * l_P^2)$$

The maximum information in a region of area A is $A/(4l_P^2)$, where l_P is the Planck length. Information is BOUNDED by area, not volume. This is the deepest constraint in physics.

6. Connections

System

Prior Chapter

Connection

<i>Puno (2026) - The Holographic Principle: 0/0 of Information</i>		
Black holes	Ch.32 Black holes	BH entropy
Entanglement	Ch.33 Entanglement	Ryu-Takayanagi
Consciousness	Ch.34 Consciousness	Holographic brain
RMT	Ch.44 RMT	SYK model
Networks	Ch.45 Networks	Information flow
Cosmic web	Ch.46 Cosmic web	Cosmological info

7. Conclusion

The holographic principle is the 0/0 of information: all information in a volume can be encoded on its boundary. $S = A/(4G_N)$ IS a 0/0: information at the horizon is neither destroyed nor preserved. The 3D world is a hologram of 2D information. This is the deepest insight in theoretical physics, connecting black holes, entanglement, consciousness, RMT, networks, and the cosmic web.

References

- [1] Bekenstein (1973), black hole entropy
- [2] Hawking (1975), particle creation by black holes
- [3] Ryu & Takayanagi (2006), holographic entanglement entropy
- [4] Maldacena (1998), AdS/CFT correspondence
- [5] Susskind (1995), holographic principle
- [6] Puno, "The Removable Singularity" (2026)